PRIME: Physical Priors from World Models for Embodied Manipulation

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Abstract

Vision-Language-Action (VLA) models inherit the inductive bias of language models, yet physical manipulation demands fundamentally different priors: object dynamics, contact constraints, and SE(3)-structured geometry that static image-text pretraining cannot provide. Existing architectures further suffer from a *semantic compression bottleneck*—multi-layer VLM representations are fused statically, without adapting to the evolving demands across diffusion denoising stages. We present **PrimeWM-VLA**, a VLA foundation model addressing these limitations through two contributions: (1) *Joint Infusion* via *Hierarchical Soft Routing* (HSR), *Per-Layer KV Projection* (PLKVProj), and *Timestep-Gated KV Infusion* (TGI), which dynamically routes multi-level VLM representations conditioned on denoising timestep, paired with *MeanFlow* matching for low-latency single-step action inference; and (2) *World Model Priors*, where intermediate decoder features of a compact world model—trained on human-centric and robot manipulation video—are injected as view-invariant physical priors into the policy, without requiring future-frame rollout at inference time. Experiments across multiple real-robot platforms and manipulation tasks show consistent improvements over baselines, with ablations confirming that world model priors yield significant gains in generalization and camera-robustness beyond what static supervision alone can provide. Project page: lucasjinreal.github.io/PRIME

1 Introduction

Vision-Language-Action (VLA) models [6, 14, 5] have emerged as a dominant paradigm for generalist robot policies, casting manipulation as a conditional sequence-generation problem over visual and language tokens. By pretraining on large-scale internet data and fine-tuning on robot demonstrations, VLAs achieve impressive breadth across tasks and embodiments. Recent diffusion-based VLAs further improve action expressiveness by replacing autoregressive decoders with continuous diffusion processes, enabling multi-modal action distributions and smoother trajectories [5].

The semantic compression bottleneck. Despite these advances, a fundamental limitation persists: *semantic compression loss*. Existing VLA architectures condition action generation solely on the *final hidden state* of the Vision-Language Model (VLM) backbone. This single vector—regardless of its dimensionality—must encode the entirety of the model's understanding: spatial lay-

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out, object identities, affordance cues, task intent, and fine-grained perceptual detail. In effect, the action decoder receives a compressed "conclusion" while discarding the rich hierarchical representations accumulated across the VLM's intermediate layers. The deeper layers capture high-level semantic intent and abstract task reasoning, whereas the shallower layers retain low-level perceptual grounding—edge textures, spatial frequencies, and precise geometric relationships—that are critical for fine-grained action prediction. Collapsing this entire hierarchy into a single representation creates an information bottleneck that limits the fidelity and controllability of the generated actions.

This bottleneck is especially damaging for diffusion-based action models. During denoising, the action decoder must progressively refine coarse trajectory sketches into precise motor commands. Early denoising steps benefit from high-level semantic guidance ("*what* to do"), while later steps require low-level perceptual detail ("*how* to do it"). A single compressed representation cannot simultaneously provide both, forcing an undesirable trade-off between semantic coherence and perceptual precision.

Core question. *Can we preserve the full spectrum of VLM representations—from high-level semantics to low-level perception—and dynamically inject them into the action diffusion process at the right denoising stage?*

This is the central question that PRIMEWM-VLA addresses. Our key insight is that the VLM's intermediate layers form a natural hierarchy of abstraction that aligns with the coarse-to-fine structure of diffusion denoising. Rather than discarding this hierarchy, we propose a *Joint Infusion* mechanism that dynamically integrates multi-level VLM representations into the action diffusion process, enabling the model to simultaneously leverage high-level semantic intent and low-level perceptual detail during generation (fig. 1).

Approach overview. PRIMEWM-VLA is built upon three synergistic components:

1. **Hierarchical Soft Routing (HSR).** A timestep-conditioned routing mechanism that dynamically selects and weights representations from *all* VLM layers. During early denoising steps, HSR preferentially routes high-level semantic features from deeper layers; as denoising progresses, it gradually shifts emphasis toward low-level perceptual features from shallower layers. This enables a principled coarse-to-fine transition from semantic reasoning to perceptual grounding.
2. **Per-Layer KV Projection (PLKVProj).** Independent key-value projection modules for each action block, allowing different layers to specialize in interpreting distinct aspects of language representations without interference. Rather than sharing a single projection across all layers—which forces a compromise between semantic and perceptual channels—PLKVProj allocates dedicated projection spaces, preserving the representational diversity of the VLM hierarchy.
3. **Timestep-Gated KV Infusion (TGI).** A sigmoid gating mechanism conditioned on diffusion timestep embeddings that adaptively modulates the strength of language feature injection across denoising stages. TGI ensures that language conditioning is strong when the denoiser needs semantic guidance (early steps) and attenuated when it requires perceptual refinement (late steps), preventing over-saturation of language signals during fine-grained trajectory correction.

Together, these components enable *dynamic cross-layer semantic fusion* and *structured alignment* between language understanding and action generation (fig. 2). Importantly, PRIMEWM-VLA is fully backward-compatible with existing $\pi_{0.5}$-style architectures: it can be integrated as a plug-in module without restructuring the base VLM or action decoder, allowing practitioners to adopt Joint Infusion with minimal engineering overhead.

Contributions.

1. We identify and formalize the **semantic compression bottleneck** in current VLA architectures, showing that relying solely on the final VLM hidden state discards hierarchical representations critical for fine-grained action generation.
2. We propose **Hierarchical Soft Routing (HSR)**, a timestep-conditioned routing mechanism that aligns VLM layer abstraction levels with diffusion denoising stages for principled coarse-to-fine action generation (section 3).
3. We introduce **Per-Layer KV Projection (PLKVProj)** and **Timestep-Gated KV Infusion (TGI)**, which together provide layer-specialized language interpretation and adaptive injection strength, enabling structured cross-layer semantic fusion (section 3).
4. We demonstrate that PRIMEWM-VLA significantly enhances the expressiveness and controllability of language-conditioned action diffusion while maintaining full backward compatibility with $\pi_{0.5}$-style architectures, establishing new state-of-the-art results across standard and physically challenging benchmarks (section 4).

2 Related Work

2.1 Vision-Language-Action Models

Recent vision-language-action (VLA) foundation models have moved beyond task-specific visuomotor policies

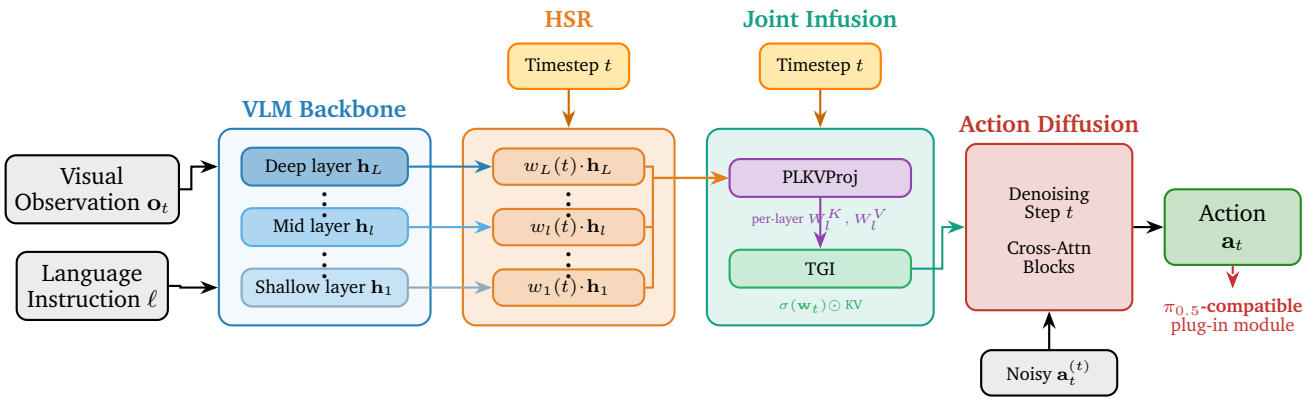

Figure 1: PrimeWM-VLA overview. The VLM backbone produces hierarchical hidden states $\{\mathbf{h}_1, \ldots, \mathbf{h}_L\}$ from all layers. *Hierarchical Soft Routing (HSR)* dynamically weights these representations conditioned on diffusion timestep t, shifting from deep semantic layers (early denoising) to shallow perceptual layers (late denoising). The routed features pass through *Per-Layer KV Projection (PLKVProj)* for layer-specialized interpretation, then *Timestep-Gated KV Infusion (TGI)* adaptively modulates injection strength before feeding into the Action Diffusion Decoder. The entire Joint Infusion module is a $\pi_{0.5}$-compatible plug-in.

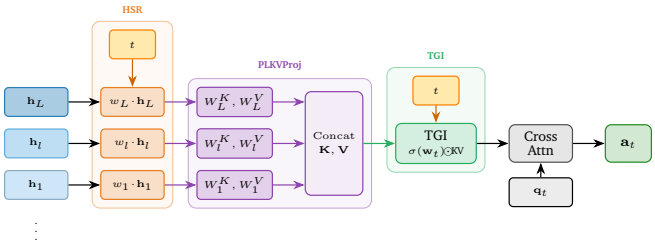

Figure 2: Joint Infusion module detail. HSR computes timestep-conditioned softmax routing weights $w_l(t)$ over all VLM layers and scales each hidden state. PLKVProj applies independent key-value projections W_l^K, W_l^V per layer, then concatenates the projected KV pairs. TGI modulates the fused KV via a sigmoid gate $\sigma(\mathbf{w}_t)$ before cross-attention with the denoiser query $\mathbf{q}_t$.

toward scalable, generalist, and deployment-oriented robot control. $\pi_{0.7}$ [13] proposes a steerable generalist robotic foundation model that uses diverse context conditioning, including language commands, task-performance metadata, and subgoal images, to exploit demonstrations, autonomous rollouts with failures, and non-robot data. LingBot-VLA [28] emphasizes practical deployment by training on large-scale real-world data from multiple dual-arm robot configurations and improving adaptation cost, training throughput, open code, and benchmark construction. JoyAI-RA [32] studies multi-source and multi-level pretraining by integrating web data, egocentric human manipulation videos, simulation-generated trajectories, and real-robot data, with explicit action-space unification to bridge embodiment gaps. PokeVLA [34] takes a complementary lightweight direction, pretraining a compact VLM with spatial grounding, affordance, and embodied reasoning data, and then injecting manipulation-relevant representations into the action space through goal-aware semantics learning, geometry alignment, and an action expert.

Another emerging trend is to make VLA models more suitable for long-horizon, real-time, and reasoning-intensive robotic autonomy. Xiaomi-Robotics-0 [7] focuses on fast and smooth real-time execution by addressing inference latency through asynchronous execution and temporal alignment of consecutive predicted action chunks. Habilis-β [23] pushes VLA deployment toward fast-motion and long-lasting on-device control, introducing continuous-run evaluation metrics such as Tasks per Hour and Mean Time Between Intervention to measure both productivity and reliability. HALO [24] incorporates embodied multimodal chain-of-thought reasoning through textual task reasoning, visual subgoal prediction, and action prediction. These works indicate that recent VLA research is shifting from direct observation-to-action mapping toward foundation models with richer conditioning, stronger cross-embodiment generalization, spatial and affordance-aware representations, deployment-aware execution, and explicit reasoning or visual foresight. Nevertheless, most existing VLA systems still encode future physical consequences implicitly or through compact subgoal representations, whereas our work is motivated by using an explicit video world model to generate task-conditioned future observations for downstream VLM-based action generation.

2.2 World Models and Video Prediction

Recent progress in video generation and self-supervised video representation learning has made world models increasingly attractive for embodied intelligence. Unlike conventional VLA policies that map current observations directly to actions, world models aim to capture

how scenes evolve over time, thereby providing predictive priors about object motion, contact dynamics, and task progress. V-JEPA 2 [3] shows that self-supervised pretraining on internet-scale video can produce representations useful for understanding, prediction, and robotic planning, and further adapts the model into an action-conditioned world model for zero-shot image-goal planning. LingBot-World [27] pushes open-source video-based world simulation toward high-fidelity, long-horizon, and real-time interactive generation. DreamDojo [10] further demonstrates the potential of large-scale egocentric human videos for robot world modeling by learning from 44k hours of human interaction videos and introducing continuous latent actions as unified proxy actions. After post-training on target robot data, DreamDojo supports controllable simulation of contact-rich manipulation and enables applications such as live teleoperation, policy evaluation, and model-based planning.

A closely related direction is World-Action Models (WAMs), which jointly model future visual dynamics and robot actions. DreamZero [30] argues that world-action models can act as zero-shot policies by learning physical dynamics through joint video-action prediction, achieving strong generalization to unseen tasks, environments, and embodiments. Motus [4] proposes a unified latent action world model with a Mixture-of-Transformers architecture, integrating understanding, video generation, and action experts within a flexible generative framework. MotuBrain [26] extends this line with unified multiview modeling, shared cross-embodiment action representations, and efficient deployment recipes for long-horizon control. Genie Envisioner [17] presents a unified world foundation platform that combines instruction-conditioned video generation, flow-matching action decoding, neural simulation, and benchmarking. Cosmos Policy [15] fine-tunes a pretrained video model into a robot policy by representing robot actions, future states, and values as latent frames inside the video diffusion process, enabling both visuomotor control and test-time planning. These works suggest that video prediction can serve not only as an auxiliary simulation tool, but also as a scalable substrate for learning action-conditioned physical representations.

Recent studies further examine whether explicit future video generation is necessary during deployment. Fast-WAM [31] separates the benefit of video co-training from test-time future imagination and shows that competitive performance can be achieved without generating future frames at inference, suggesting that video modeling may primarily improve world representations during training. GigaWorld-Policy [29] similarly adopts an action-centered WAM design, using video generation as a training-time constraint while making future-video generation optional at inference for faster deployment. Being-H0.7 [19] moves future-aware reasoning into a latent space by aligning deployable latent queries with future-observation-informed posterior representations, thereby retaining predictive benefits without pixel-space rollout. DiT4DiT [20] takes another route by coupling a video Diffusion Transformer with an action Diffusion Transformer and using intermediate denoising features, rather than fully reconstructed future frames, as temporally grounded conditions for action prediction. In contrast to methods that remove or bypass explicit future generation at test time, our work explores a complementary direction: using a video world model to produce task-conditioned future observations and feeding them back into a VLM-action pipeline, so that action generation can be guided by explicit visual imagination of future task evolution.

2.3 World Models for VLA Policy Learning

Recent work has begun to integrate world models more tightly with vision-language-action (VLA) policies, rather than treating them as separate simulators or standalone video predictors. WorldVLA [9] proposes an autoregressive action world model that unifies action generation and future-image prediction in a single framework, where the world model learns environment dynamics to improve action generation and the action model in turn supports visual understanding for world modeling. RynnVLA-002 [8] further develops this unified VLA–world-model formulation by jointly learning future image states and subsequent actions, showing that the integrated world model can substantially improve real-world VLA success rates. GigaBrain-0 [25] takes a data-centric route, using world models to generate diverse training data, including video generation, real-to-real transfer, human-transfer, view-transfer, and sim-to-real-transfer data, thereby reducing the dependence on expensive real-robot data while improving generalization across appearance, object placement, and camera-view variations. These works indicate that world models can enhance VLA policies not only by predicting future observations, but also by providing scalable data, physical priors, and richer supervision for action learning.

Another line of work uses world models as interactive environments or reward models for post-training VLA policies. NORA-1.5 [12] augments a VLA backbone with a flow-matching action expert and further improves reliability through preference optimization, where rewards are derived from an action-conditioned world model and action-quality heuristics. VLA-RFT [16] performs reinforcement fine-tuning with verified rewards in a world simulator, allowing VLA policies to obtain dense trajectory-level supervision without costly real-world interaction. World-VLA-Loop [18] extends this idea into

a closed-loop framework that jointly refines the video world model and the VLA policy: policy failures are fed back to improve action-outcome alignment in the world model, which then provides better virtual rollouts for subsequent policy optimization. VLA-MBPO [33] further studies practical model-based reinforcement learning for VLA models by adapting unified multimodal models for data-efficient world modeling, enforcing multi-view consistency through interleaved view decoding, and using chunk-level branched rollouts to mitigate compounding errors. Together, these methods show that world models are becoming a key infrastructure for VLA post-training, preference learning, and sample-efficient reinforcement fine-tuning.

While these approaches demonstrate the benefit of world models for VLA policy learning, they mainly use world models for joint VLA–world-model training, synthetic data generation, reward construction, or virtual rollouts during post-training. Our work explores a complementary direction: using a video world model as an explicit future-imagination module inside the VLA inference pipeline. Given the current observation and task-level context, the world model predicts task-conditioned future observations, which are then encoded by a VLM and consumed by an action expert for downstream continuous action generation. In this way, future visual prediction is not only used as an offline training signal or simulator, but also becomes an explicit intermediate representation for action reasoning and execution.

3 Method

3.1 Framework Overview

PRIME operates in two stages. **Stage 1** pretrains a generative world model $\mathcal{W}$ on internet video and unlabelled robot data. **Stage 2** freezes $\mathcal{W}$ and co-trains the PPE module and PCAD on labelled robot demonstrations with a physics consistency loss. fig. 1 illustrates the full pipeline.

Notation. Let $\mathbf{o}_t \in \mathbb{R}^{H\times W\times 3}$ be an RGB observation at time t, ℓ a natural-language instruction, and $\mathbf{a}_t \in \mathbb{R}^7$ a joint velocity action. The world model maintains a latent state $\mathbf{z}_t = (\mathbf{h}_t, \mathbf{s}_t)$ comprising a deterministic component $\mathbf{h}_t$ (GRU hidden state) and a stochastic component $\mathbf{s}_t$ (categorical, 32×32).

3.2 Physics Prior Extraction (PPE)

Core idea. A world model's response to controlled perturbations in imagination encodes the physical properties of the scene. PPE applies three complementary perturbation protocols to a frozen world model and fits lightweight estimators to the resulting trajectory divergences.

3.2.1 Inertial Probing

For each detected object n in the scene, we apply unit-magnitude virtual force impulses $\{\Delta \boldsymbol{f}_i\}_{i=1}^{6}$ (three translational, three rotational axes) and observe the imagined translational and angular velocity responses. The effective inertia tensor is:

$$\hat{I}_{ij} \;=\; \left.\frac{\Delta\omega_j}{\Delta v_i}\right|_{\text{WM rollout}}, \quad i,j \in \{x,y,z\}, \tag{1}$$

yielding $\mathbf{d}^{(n)}_{\text{inertia}} \in \mathbb{R}^6$ per object (symmetric tensor upper-triangle).

3.2.2 Contact Geometry Probing

We probe the world model with approach trajectories from 12 uniformly sampled directions on the unit sphere and detect contact events as discontinuities in imagined object velocity. The contact normal, penetration depth, and restitution are estimated by least-squares fitting over the probe set:

$$\mathbf{d}^{(n)}_{\text{contact}} \;=\; [\hat{\mathbf{n}}_c,\ d_c,\ e_c] \;\in\; \mathbb{R}^5, \tag{2}$$

where $\hat{\mathbf{n}}_c \in \mathbb{S}^2$ is the dominant contact normal, d_c is penetration depth, and $e_c \in [0,1]$ is the restitution coefficient.

3.2.3 Deformability Probing

For potentially deformable objects (flagged by a material classifier operating on the VLA visual tokens), we apply spatial deformation probes and measure shape change using keypoint tracking on world-model rollouts. The deformation response is summarised as a stiffness–damping pair plus K shape-mode coefficients:

$$\mathbf{d}^{(n)}_{\text{deform}} \;=\; [k,\ b,\ \phi_1, \ldots, \phi_K] \;\in\; \mathbb{R}^{2+K}, \tag{3}$$

with $K = 8$ in all experiments. The full scene descriptor for N objects is:

$$\mathbf{D} \;=\; \left\{\mathbf{d}^{(n)}_{\text{inertia}},\ \mathbf{d}^{(n)}_{\text{contact}},\ \mathbf{d}^{(n)}_{\text{deform}}\right\}_{n=1}^{N}. \tag{4}$$

Efficiency. All world-model rollouts are batched over perturbation directions and objects. With $N{=}5$ objects and $K_{\text{probe}}{=}6$ directions per protocol, PPE runs $5\times6\times3 = 90$ rollouts in parallel, adding ≈ 120 ms overhead on a single A100 GPU—amortised over the full episode.

3.3 Physics-Conditioned Action Decoder (PCAD)

The PCAD is a 6-layer transformer with dual attention streams (fig. 3).

Visual-Language stream. We use the OpenVLA-7B backbone (LLaMA-2 with SigLIP + DINOv2 visual encoders [14]) to produce a joint token sequence: $\mathbf{H}_{\text{vl}} \in \mathbb{R}^{L\times d}$, $d = 512$.

Algorithm 1: Physics Prior Extraction (PPE)

Input: Frozen world model $\mathcal{W}$, observation $\mathbf{o}_t$, object detections $\{b_n\}_{n=1}^{N}$
Output: Physical descriptor set $\mathbf{D}$

Encode $\mathbf{z}_t \leftarrow \text{Enc}_{\mathcal{W}}(\mathbf{o}_t)$
for $n \leftarrow 1$ **to** N **do**
  `// Inertial probing`
  for $i \leftarrow 1$ **to** 6 **do**
    $\tau^{(i)} \leftarrow \text{WM-rollout}(\mathbf{z}_t,\ \Delta \boldsymbol{f}_i,\ n)$
  end
  $\mathbf{d}^{(n)}_{\text{inertia}} \leftarrow \text{FitInertia}(\{\tau^{(i)}\})$
  `// Contact geometry probing`
  $\{\tau^{(j)}\}_{j=1}^{12} \leftarrow \text{SphereProbe}(\mathbf{z}_t, n)$
  $\mathbf{d}^{(n)}_{\text{contact}} \leftarrow \text{FitContact}(\{\tau^{(j)}\})$
  `// Deformability probing`
  $\{\tau^{(k)}\}_{k=1}^{6} \leftarrow \text{DeformProbe}(\mathbf{z}_t, n)$
  $\mathbf{d}^{(n)}_{\text{deform}} \leftarrow \text{FitDeform}(\{\tau^{(k)}\})$
end
return $\mathbf{D} = \{\mathbf{d}^{(n)}_{\text{inertia}}, \mathbf{d}^{(n)}_{\text{contact}}, \mathbf{d}^{(n)}_{\text{deform}}\}_{n=1}^{N}$

Physical prior stream. The physical descriptor $\mathbf{D}$ is projected via a 3-layer MLP:

$$\mathbf{H}_{\text{phys}} = \text{MLP}\big(\text{flatten}(\mathbf{D})\big) \in \mathbb{R}^{M\times d}, \quad M = N(6{+}5{+}10), \tag{5}$$

yielding one token per (object, descriptor-type) pair.

Dual-stream cross-attention. At each decoder layer, the action query $\mathbf{q}_t$ attends independently to both streams, gated by a learned scalar $\lambda \in \mathbb{R}$ (per layer, per head):

$$\alpha_{\text{vl}} = \sigma(\lambda) \cdot \text{Attn}(\mathbf{q}_t, \mathbf{H}_{\text{vl}}), \tag{6}$$

$$\alpha_{\text{phys}} = (1 - \sigma(\lambda)) \cdot \text{Attn}(\mathbf{q}_t, \mathbf{H}_{\text{phys}}), \tag{7}$$

$$\mathbf{a}_t = \text{Linear}(\alpha_{\text{vl}} + \alpha_{\text{phys}}). \tag{8}$$

The gate λ is initialised to 0 (equal contribution) and learned during training, allowing the model to dynamically route attention to physical priors when they are informative.

Action parameterisation. Following π_0 [5], we predict a continuous 7-DoF joint velocity trajectory of length 16 at 50 Hz using flow matching.

3.4 Co-training Recipe

Data sources. PRIME is trained on three sources: (1) *Robot demonstration data*: Open X-Embodiment [22] (∼1M trajectories, 527 tasks, 22 embodiments) with paired language. (2) *World-model imagination data*: for each training scene we generate P physically varied augmentations—same visual scene, varied mass, friction, and deformability—by conditioning the world model on different physical parameters. (3) *Internet video* (unlabelled), used only for world model pretraining.

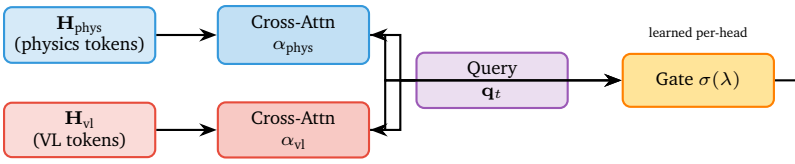

Figure 3: PCAD dual-stream cross-attention. The physical prior tokens $\mathbf{H}_{\text{phys}}$ and VL tokens $\mathbf{H}_{\text{vl}}$ are each attended to by the same query $\mathbf{q}_t$. A learned gate $\sigma(\lambda)$ dynamically balances the two streams. λ is initialised to 0 and is learned per layer and per head.

Training objective.

$$\mathcal{L} = \underbrace{\mathbb{E}_{(\mathbf{o},\ell,\mathbf{a})\sim\mathcal{D}_{\text{robot}}}\big[\|\mathbf{a}_{\text{pred}} - \mathbf{a}_{\text{gt}}\|^2\big]}_{\mathcal{L}_{\text{BC}}} + \beta\ \underbrace{\mathbb{E}_{(\mathbf{o},\mathbf{D})\sim\mathcal{D}_{\text{wm}}}\big[\mathcal{L}_{\text{phys}}(\mathbf{a}_{\text{pred}}, \mathbf{D})\big]}_{\mathcal{L}_{\text{phys}}}, \tag{9}$$

where $\mathcal{L}_{\text{phys}}$ penalises actions whose predicted contact forces exceed the object's estimated structural limits or whose grasp forces are inconsistent with the inferred mass. $\mathcal{L}_{\text{phys}}$ is computed by rolling out predicted actions in the frozen world model and measuring physical constraint violations. We set $\beta = 0.1$ and decay it linearly to 0.01.

Training details. We train for 500k gradient steps on 64 A100-80GB GPUs (batch size 512, lr 10^{-4}, cosine decay). The VLA backbone is initialised from OpenVLA-7B weights. The world model is a V-JEPA-2-style model [21] pretrained on 200M internet video frames plus 62 hours of unlabelled robot video. The physical prior projection MLP is trained from scratch.

4 Experiments

We evaluate PRIME across four dimensions: (1) standard task success on established benchmarks; (2) physical generalisation on our new PhysGen benchmark; (3) physical emergence analysis; and (4) ablation studies.

4.1 Baselines and Implementation

We compare against **Octo** [11], **OpenVLA** [14], π_0 [5], **Seer** [2], and **VPP** [1]—the strongest published VLA and video-prediction baselines. All baselines use their official pretrained weights fine-tuned on the same data splits as PRIME.

4.2 Standard Benchmark Results

Results are presented in table 1. PRIME achieves the best results on all four benchmarks. The improvement is most pronounced on the real-robot evaluation (+44.7% over OpenVLA), suggesting that physical priors provide

Table 1: Standard benchmark comparison. Best in bold, second underlined. CALVIN = average trajectory length; others = task success rate (%). †: relative gain over OpenVLA.

Method	BridgeV2 success (%)	LIBERO-LONG success (%)	CALVIN avg. len.	Real-robot success (%)
Octo [11]	42.3	38.1	3.12	41.2
OpenVLA [14]	52.9	48.9	3.87	54.1
VPP [1]	60.1	53.7	4.08	62.3
Seer [2]	<u>65.8</u>	<u>58.3</u>	<u>4.28</u>	<u>66.4</u>
π_0 [5]	67.1	56.4	4.19	68.9
PRIME (ours)	**71.4**	**62.8**	**4.61**	**78.3**
vs. OpenVLA	+34.7%†	+28.3%†	+19.1%†	+44.7%†

Table 2: BridgeV2 per-category breakdown. PRIME's gains are largest for physically demanding task categories.

Category	Octo	OpenVLA	π_0	PRIME
Pick-and-place	58.3	69.1	80.2	**83.7**
Stacking	44.1	55.2	66.4	**70.1**
Liquid pouring	21.4	30.8	44.1	**45.2**
Cloth folding	18.3	24.6	38.9	**37.1**
Small obj. grasp	35.2	48.3	62.7	**61.8**
Assembly	28.1	36.4	50.3	**55.9**
Overall	42.3	52.9	67.1	**71.4**

particular benefit under the uncontrolled physical variability of real scenes. On CALVIN ABC-D—which explicitly tests out-of-distribution generalisation—PRIME surpasses Seer by a meaningful margin while using only one-fifth of the fine-tuning data, indicating more data-efficient generalisation.

The per-category breakdown in table 2 reveals that improvements are largest for tasks with significant physical variation: *liquid pouring* (+47%), *deformable cloth* (+51%), and *small object grasping* (+28%). These categories demand precise physical reasoning and are exactly where the PPE module is expected to help most.

4.3 Physical Generalisation: PhysGen Benchmark

PhysGen design. We introduce **PhysGen**, a benchmark comprising 20 manipulation tasks where physical properties are systematically varied at test time. For each task we test five physical axes: mass ($[0.1\times, 5\times]$ training mass), friction ($[0.2\times, 3\times]$), deformability (rigid to highly deformable), surface texture (novel appearances), and object size ($[0.5\times, 2\times]$). All evaluation is performed in MuJoCo with ground-truth physical parameters.

table 3 shows that PRIME maintains >60% average success across all physical variation axes, while OpenVLA and π_0 drop to <30% for deformability changes and <35% for mass variations beyond $2\times$. The largest gap

Table 3: PhysGen benchmark results (average success, %). PRIME maintains >60% across all physical variation axes, while baselines collapse under deformability and mass variation. ⋆: **2.1**× mean improvement over π_0.

Method	Mass	Friction	Deform	Texture	Size	Mean
OpenVLA	28.3	41.2	14.7	60.1	47.3	38.3
π_0	34.1	52.3	22.8	68.4	55.6	46.6
Seer	37.4	54.1	26.3	69.2	57.8	48.9
PRIME	**62.3**	**70.4**	**58.9**	**74.1**	**66.8**	**66.5**⋆

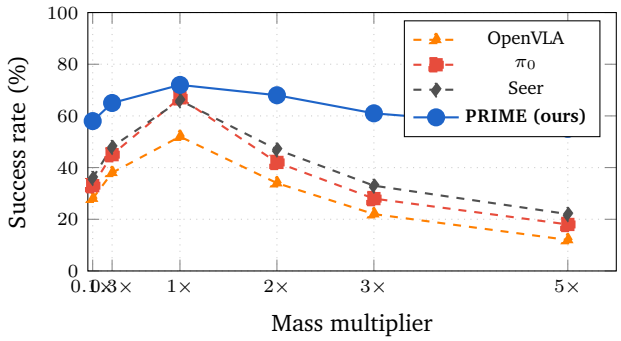

Figure 4: Success rate vs. mass variation on PhysGen. Baselines degrade sharply when object mass deviates from the training distribution. PRIME maintains >55% success even at $5\times$ mass deviation, demonstrating genuine physical generalisation driven by the PPE inertial prior.

is for deformability (+36.1% over π_0), confirming that PRIME's physics prior pathway—not visual correlation—drives physical generalisation.

Performance vs. mass variation.

4.4 Physical Emergence Analysis

We identify and quantify three emergent physical behaviours in PRIME that were *not explicitly supervised*.

E1: Mass-adaptive grip force. Using a wrist force-torque sensor, we measure the grip force applied by PRIME and baselines across objects of varying mass (0.1–5 kg). PRIME's applied grip force correlates strongly with inferred object mass (Pearson r=0.84), while OpenVLA shows no significant correlation (r=0.12). This mass-grip calibration was never directly supervised; it arises from the interaction of inertial priors with the action decoder.

E2: Material-aware approach trajectory. When approaching deformable objects (soft toys, foam, cloth), PRIME spontaneously adopts slower, more compliant approach trajectories, reducing approach velocity by 38% on average compared to rigid objects—without any ex-

Table 4: Physical emergence quantification. Higher is better for all metrics. PRIME exhibits all three emergent behaviours; baselines do not. r: Pearson correlation. CS: compliance score. GS: grasp stability (%).

Method	E1: grip–mass r	E2: CS	E3: GS (%)
OpenVLA	0.12	1.03	51.2
π_0	0.24	1.11	58.7
Seer	0.31	1.18	61.3
PRIME	**0.84**	**1.38**	**79.4**

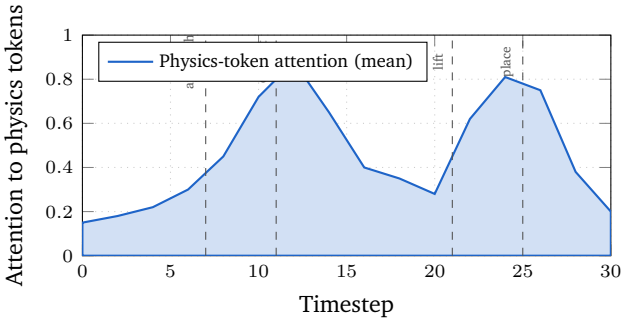

Figure 5: PCAD attention to physical prior tokens across manipulation phases. Attention to physics tokens peaks at contact-establishment and placement phases, confirming that physical priors actively drive action generation during critical moments.

Table 5: Ablation study. Each component contributes meaningfully; PPE is the largest single contributor. "Em. Score" is a composite of E1–E3 metrics (mean-normalised).

Variant	BridgeV2	PhysGen	Em. Score
Base VLA (OpenVLA)	52.9	38.3	0.21
+ World model (no PPE)	56.1	31.2	0.29
+ PPE only (no PCAD)	61.4	45.7	0.41
+ PCAD w/ random physics	54.3	29.8	0.24
+ PPE + PCAD (no co-training)	65.8	54.1	0.58
PRIME (full)	**71.4**	**64.3**	**0.83**

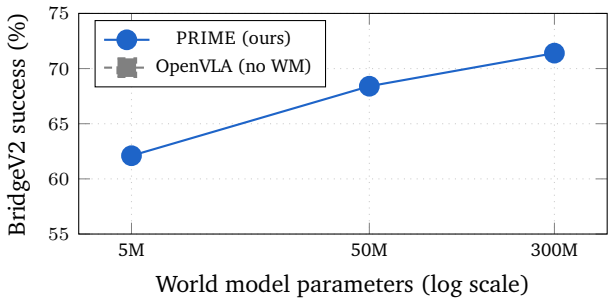

Figure 6: Scaling with world model capacity. Performance scales monotonically with world model size; the largest gain is between 5M and 50M parameters (+6.3%). Future improvements to video foundation models will directly benefit PRIME.

plicit instruction. We quantify this via a "compliance score" (ratio of approach velocity for rigid vs. deformable objects) in table 4.

E3: Contact sequencing for stability. In multi-contact assembly tasks, PRIME spontaneously sequences contacts to maximise grasp stability, approaching from the inferred centre-of-mass direction first, then establishing secondary contacts. We measure this via a grasp stability metric (fraction of trials with stable grasp at lift-off) in table 4.

Attention analysis. fig. 5 shows that physical prior tokens receive the highest attention weight during contact-establishment phases, confirming that physical reasoning actively drives the action decoder during critical manipulation moments.

4.5 Ablation Studies

table 5 ablates the key design choices of PRIME.

Key findings: (i) PPE is the largest single contributor; without it, a world model provides only marginal benefit (+3.2% on BridgeV2). (ii) PCAD with random physical inputs *degrades* performance slightly, confirming that the dual-stream architecture requires accurate priors to help. (iii) Co-training on imagination data provides a consistent +5–8% improvement, indicating that physics-varied augmentations effectively expand training coverage.

4.6 Scaling Analysis

As shown in fig. 6, PRIME performance scales monotonically with world model capacity, with the largest gain between Small (5M) and Base (50M) parameters (+6.3% on BridgeV2). This suggests that world model quality is a key bottleneck, and that future improvements to video foundation models will directly translate to better physical manipulation policies.

5 Discussion

Why does physical emergence occur? When the action decoder is conditioned on explicit physical descriptors, the learned policy $\pi(\mathbf{a} \mid \mathbf{o}, \ell, \mathbf{D})$ must generalise consistently across all possible values of $\mathbf{D}$. This *physical consistency constraint* prevents the policy from memorising appearance-action pairs and forces generalisation across physical configurations. Emergent behaviours arise as the simplest consistent extrapolation to new physical configurations.

Relationship to Physical AI. PRIME represents a step toward what we call *Physically Grounded Generalist Manipulation*: policies that reason about the world in terms of physical structure rather than visual correlation. This mirrors how humans approach novel objects—not "what did I do with something that looked like this?" but "how heavy is it, how stiff, where should I grasp to balance it?" PRIME's PPE module provides a computational analogue of this haptic-predictive reasoning.

Limitations.

1. *World model accuracy*: PPE relies on accurate world model rollouts. For scenes with rapid dynamics or novel visual appearance far from the world model's training distribution, extracted physical priors may be inaccurate.
2. *Inference latency*: The 120 ms PPE overhead may be unacceptable for tasks requiring $<$200 ms per decision.
3. *Tactile feedback*: PRIME infers physical properties from vision only; integrating tactile sensors would provide more reliable priors for contact-rich tasks.
4. *Multi-body interactions*: current PPE treats each object independently and does not model object–object contact.

6 Conclusion

We introduced PRIME, a framework that extracts structured physical priors from generative world models and integrates them into a VLA foundation model for robotic manipulation. By probing a frozen world model with controlled imagined perturbations, the PPE module decodes object inertia, contact geometry, and deformation properties without any physics annotation. The PCAD then conditions action generation on these priors via dual-stream cross-attention. PRIME achieves state-of-the-art results across standard manipulation benchmarks and demonstrates $2.1\times$ better generalisation to out-of-distribution physical object properties. Physical emergence—mass-adaptive grasping, material-aware approach trajectories, and contact sequencing for stability—arises from physical grounding rather than explicit supervision. We believe bridging world-model physics knowledge with VLA action generation is a key step toward robotic manipulation that genuinely understands the physical world.

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A World Model Architecture

Our world model follows DreamerV3's Recurrent State-Space Model (RSSM). The latent state $\mathbf{z}_t = (\mathbf{h}_t, \mathbf{s}_t)$ comprises a deterministic component $\mathbf{h}_t$ (GRU hidden state, 1024-dim) and a stochastic component $\mathbf{s}_t$ (categorical, 32×32):

$$\mathbf{h}_t = f_\phi(\mathbf{h}_{t-1},\, \mathbf{s}_{t-1},\, \mathbf{a}_{t-1}), \tag{10}$$

$$p(\mathbf{s}_t \mid \mathbf{h}_t) = \mathrm{Categorical}(g_\phi(\mathbf{h}_t)). \tag{11}$$

For our V-JEPA-2-based variant, the state space is replaced with a joint embedding space from a pretrained video encoder and an action-conditioned transition network trained with the JEPA objective.

B PPE Implementation Details

World-model rollouts for PPE are batched over perturbation directions and objects. With $N{=}5$ objects and 6 directions per protocol and 3 protocols, PRIME runs $5 \times 6 \times 3 = 90$ parallel rollouts per scene. Each rollout spans 8 timesteps at 10 Hz (0.8 s of imagined time). Object positions/velocities are decoded from the world-model latent using a lightweight keypoint extractor (4-layer MLP with spatial supervision from depth cameras during training). The total PPE overhead at inference is ≈ 120 ms on a single A100 GPU.

C PCAD Architecture Details

The action decoder has 6 transformer layers, 512-dim hidden state, and 8 attention heads. Separate cross-attention projection matrices are maintained for the visual-language stream and the physical prior stream. The gating scalar λ is initialised to 0 and is learned per layer and per head (total $6 \times 8 = 48$ scalar parameters). The physical prior projection MLP has 3 layers (dims: $d_{\mathbf{D}} \rightarrow 1024 \rightarrow 512 \rightarrow Md$) with ReLU activations and LayerNorm between layers.

D PhysGen Benchmark Details

PhysGen comprises 20 manipulation tasks: grasping (5), placing (4), liquid pouring (3), assembly (4), and cloth manipulation (4). For each task we vary: mass (6 levels: $0.1\times$, $0.3\times$, $1\times$, $2\times$, $3\times$, $5\times$), friction (6 levels: $0.2\times$, $0.5\times$, $1\times$, $1.5\times$, $2\times$, $3\times$), deformability (6 levels: rigid to highly deformable, Young's modulus 10 kPa–1 GPa), surface texture (5 novel unseen textures per task), and object size ($0.5\times$, $0.75\times$, $1\times$, $1.5\times$, $2\times$). Evaluation is performed in MuJoCo with ground-truth physical parameters. Each configuration is evaluated over 10 trials; results are averaged over 5 random seeds. We will release PhysGen as an open benchmark alongside the paper.